

Lecture 4. Logistic Regression

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Overview

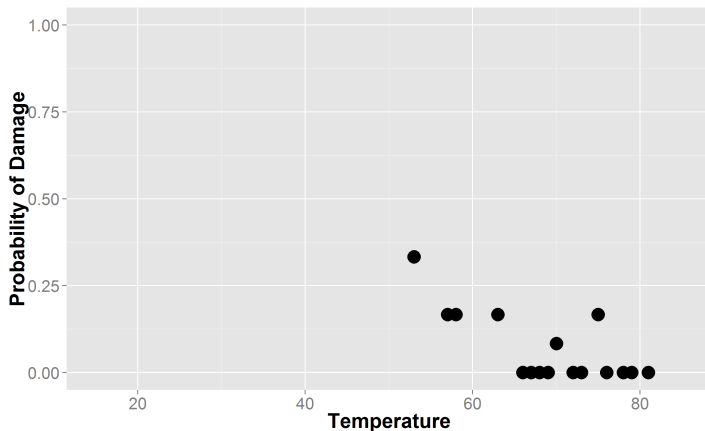
- 1 Logistic Regression
- 2 Interpretation of Coefficients
- 3 Inference

Example 1: O-rings on the Space Shuttle

- O-rings fail at higher rates in colder temperatures
- There are 6 O-rings on each shuttle
- There were 23 missions prior to Challenger
- Could the Challenger explosion have been predicted?

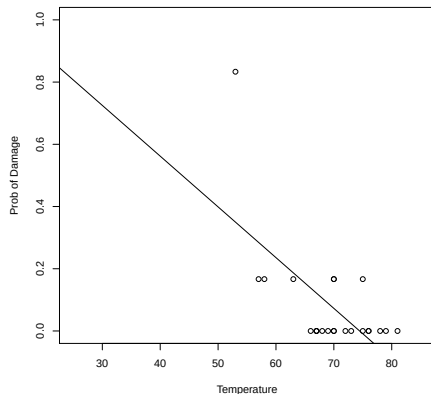
Example 1: O-rings on the Space Shuttle

First, plot the failure rates by temperature:



Example 1: O-rings on the Space Shuttle

It looks like there might be a relationship. Let's regress failure rate on temperature. Here's the resulting regression line:



Linear Model and Binary Outcome

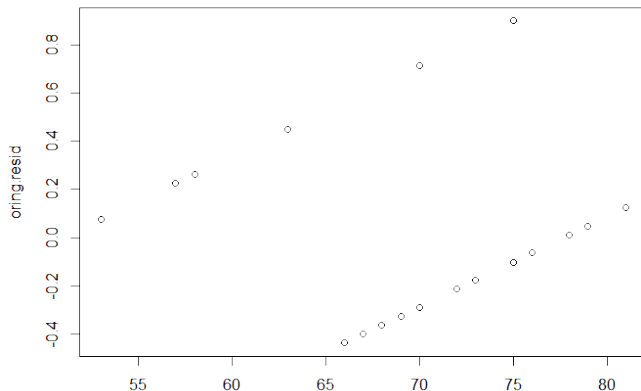
In some cases, the linear model might not produce a bad fit, but there are some issues.

- 1 Linear regression assumes residuals are normally distributed.
- 2 Linear regression also assumes constant variance.
- 3 Linear model can produce predicted values outside the restricted range of a probability $(0,1)$.

Violations of assumptions mean linear model won't always/usually be appropriate.

Linear Model and Binary Outcome

Assume that residuals are normally distributed. These are the residuals from a linear model $DAMAGE = \beta_0 + \beta_1 \times TEMP$:



Linear Model and Binary Outcome

Recall the binomial distribution:

$$P(Y_i = y_i) = \frac{n_i!}{y_i!(n_i - y_i)!} p_i^{y_i} (1 - p_i)^{n_i - y_i} \quad (1)$$

This distribution has mean $n_i p_i$ and variance $n_i p_i (1 - p_i)$. The mean and the variance are linked in this case – nonconstant. Violation of second assumption.

Linear Model and Binary Outcome

Let's say we also have q predictors $(x_{i1}, \dots, x_{iq})$. We want to use these to predict p_i .

For the reasons described earlier, that may not work.

Question:

Is it possible to define a function, $g(p_i)$ for which the linear regression model will work?

Logistic Regression

This is the basic idea behind logistic regression (and many methods under the heading of generalized linear models [GLMs]).

We need some function $\eta_i = g(p_i)$ such that η_i can be predicted in a linear model.

It must also satisfy $0 \leq g^{-1}(\eta) \leq 1$.

The most common function is called the logit function: $\eta = \log\left(\frac{p}{1-p}\right)$.

Logistic Regression

Table: Relationship of Odds and Log-odds to Probabilities

Probability	Odds	Log-Odds
0	0	$-\infty$
0.1	0.111	-2.197
0.2	0.25	-1.386
0.3	0.429	-0.847
0.4	0.667	-0.405
0.5	1	0
0.6	1.5	0.405
0.7	2.333	0.847
0.8	4	1.386
0.9	9	2.197
1	∞	∞

Logistic Regression

Logistic regression fits the following model:

$$\begin{aligned}\eta &= \beta_0 + \beta_1 x_{i1} + \dots + \beta_q x_{iq} \\ \log\left(\frac{p}{1-p}\right) &= \beta_0 + \beta_1 x_{i1} + \dots + \beta_q x_{iq} \\ Pr(Y_i = 1) &= \text{logit}^{-1}(\beta_0 + \beta_1 x_{i1} + \dots + \beta_q x_{iq}) \\ Pr(Y_i = 1) &= \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{i1} + \dots + \beta_q x_{iq})}}\end{aligned}$$

Logistic Regression

The R code is straightforward. We'll start with the basics and add to this as we go:

```
logitmod<-glm(damage ~temp, family=binomial,  
data=spacshu)
```

Logistic Regression

If we fit the model $\eta = \beta_0 + \beta_1 x_{i1}$, where $\eta = \log\left(\frac{p}{1-p}\right)$ and x is the temperature, we get the following:

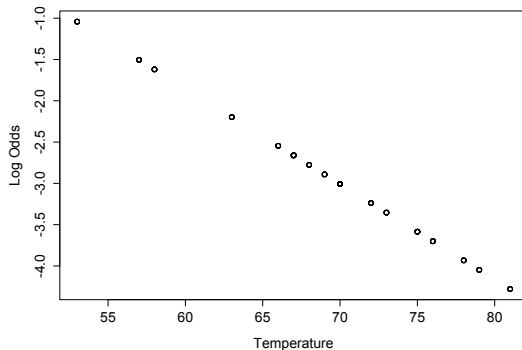
Coefficients:

(Intercept)	temp
5.0850	-0.1156

In this case $\eta = 5.0850 - 0.1156 * TEMP$.

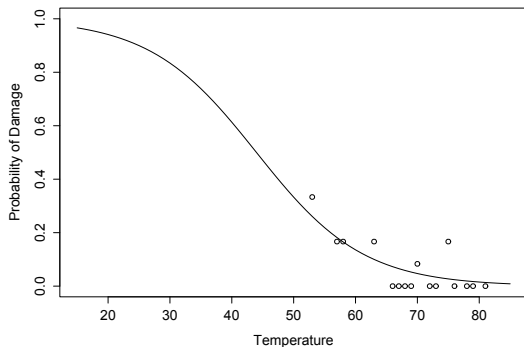
Logistic Regression

Let's plot the predicted values from this model.



Logistic Regression

We could also convert this to the probability scale:



Logistic Regression

It is possible to use many of the techniques that you learned last semester to predict the logit in this setup.

- Transformations
- Dummy variables
- Interactions

Interpretation of Coefficients

But, we are working on the scale of the logit. This means that the results need careful interpretation.

Recall the $\frac{p}{1-p}$ is called the odds. So η is actually a log of the odds.

We can write our model:

$$\log(\text{odds}) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_q x_{iq}$$

Interpretation of Coefficients

$$\log(\text{odds}) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_q x_{iq}$$

$$\text{odds} = \exp\{\beta_0 + \beta_1 x_{i1} + \dots + \beta_q x_{iq}\}$$

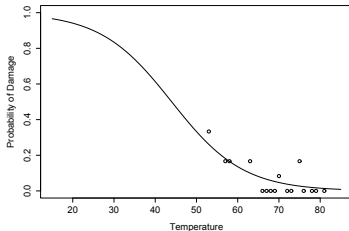
In this case, β_1 can be interpreted as meaning a 1-unit increase in x_{i1} with all other x 's being held constant increases the log-odds of success by β_1 .

Therefore, e^{β_1} can be interpreted as the increase in the odds-ratio that would result from a 1-unit increase in x_{i1} with all other x 's being held constant.

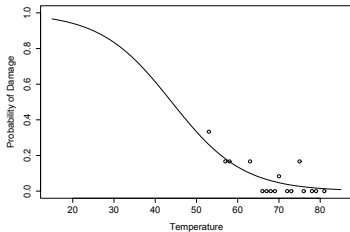
Interpretation of Coefficients

One issue is that people want to interpret results on the probability scale. This won't work very easily with logistic regression.

Recall that the inverse-logistic function is curved:



Interpretation of Coefficients



The steepest change occurs at the middle of the curve.

Interpretation of Coefficients

Example Points

- $\text{logit}(0.50)=0$ and $\text{logit}(0.60)=0.4$. At this point, adding 0.4 on the logit scale increases the probability 10%.
- $\text{logit}(0.90)=2.2$ and $\text{logit}(0.93)=2.6$. At this point, adding 0.4 on the logit scale increases the probability 3%.

Interpretation of Coefficients

One method is to evaluate at the mean of the predictor variable of interest.

In the O-ring example, the mean temperature is 69.57. Therefore, the probability of O-ring failure at that temperature is

$$\frac{1}{1 + e^{-(5.0850 - 0.1156 * 69.57)}} = 0.05.$$

Interpretation of Coefficients

Another method is to evaluate at several points. This assumes that all the values of all other predictors are held constant (if we had more x 's).

Table: Pr(Failure) at Different Temperatures

Temperature	Logit	Pr(Failure)	Increase
69.57	-2.96	0.05	
59.57	-1.80	0.17	0.11
49.57	-0.65	0.52	0.36

Interpretation of Coefficients

The 'divide by 4 rule'.

The slope of the curve is steepest at its center point. At this point, $\beta_0 + \beta_1 x_{i1} = 0$ so that $\text{logit}^{-1}(0) = 0.5$. The slope of the curve at that point is $\beta_1/4$.

Therefore, the maximum difference in probability corresponding to a 1-unit increase in x is $\beta_1/4$. This maximum occurs where the probabilities are close to 0.5. In our example $-0.1156/4 = -0.0289$.

Interpretation of Coefficients

Why does the 'divide by 4 rule' work?

The slope of the curve reaches its maximum at $\beta_0 + \beta_1 * x_{i1} = 0$. At this point, $\text{logit}^{-1}(\beta_0 + \beta_1 * x_{i1}) = 0.5$.

You can use the first derivative of the logit function with respect to β_1 to get an estimate of the maximum slope:

$$\beta \frac{e^0}{(1 + e^0)^2} = \beta \frac{1}{(1 + 1)^2} = \frac{\beta}{4} \quad (2)$$

Interpretation of Coefficients

Another already mentioned interpretation of the coefficients in a logistic regression model is as odds ratios.

Recall that the log of the odds ratio has a symmetric distribution.

```
> exp(cbind(coef(logitmod), confint(logitmod)))
Waiting for profiling to be done...
              2.5 %          97.5 %
(Intercept) 161.5762606 0.3641231 7.207593e+04
temp         0.8908304 0.8087817 9.758269e-01
```

Interactions

It is possible to include interactions in logistic regression models.

These interactions need careful interpretation.

Interactions

Example from Chen (2003).

Variable (Reference)	OR
Age, y(<30)	2.5
Married	1.1
No. of children (≥ 3)	0.8
Contraceptive use	0.1
Aware of vertical transmission	1.7
Planned pregnancy	1.7
HIV infected	3.0
No. of HIV-related symptoms (≥ 2)	0.9
Intention x infected	0.1
Frequency of sexual activity	0.9
Treatment group	1.6

Interactions

Calculating odds ratios in the presence of interactions. For women who were not planning a pregnancy, the OR for those with HIV infection vs. those without:

$$\begin{aligned}\hat{OR}_1 &= \frac{\text{Odds for } (I_{planned} = 0 \text{ and } I_{infected} = 1)}{\text{Odds for } (I_{planned} = 0 \text{ and } I_{infected} = 0)} \\ &= \exp(\hat{\beta}_{infected}) \\ &= \exp(1.0986) = 3.0\end{aligned}$$

Interactions

For women with a planned pregnancy, the OR for those with HIV infection vs. those without:

$$\begin{aligned}\hat{OR}_1 &= \frac{\text{Odds for } (I_{planned} = 1 \text{ and } I_{infected} = 1)}{\text{Odds for } (I_{planned} = 1 \text{ and } I_{infected} = 0)} \\ &= \exp(\hat{\beta}_{infected} + \hat{\beta}_{planned \times infected}) \\ &= \exp(1.0986 - 2.3026) = 0.3\end{aligned}$$

Summary: Write out the OR of interest in this including both of the interacted variables and derive the OR appropriately.

Reference vs Effect/Sum Coding

There are alternative **parameterizations** for categorical predictors when used in logistic regression (and elsewhere).

These parameterizations don't change predicted values, but the estimated coefficients are different and, therefore, require different interpretations.

Two different parameterizations we will look at:

- 1 Reference coding
- 2 Effect or sum coding

Reference vs Effect/Sum Coding

The different codings are based on different **contrasts**.

Reference coding treats one category as a reference group.

This group's probability is estimated via the intercept.

Other groups have coefficients representing differences from the reference group.

Reference vs Effect/Sum Coding

Let's consider an example. The probability of a low birth weight baby.

Using `RACE` with three categories as a predictor.

The reference coding for this:

```
> contrasts(lowbwt$RACE)
      2  3
1  0  0
2  1  0
3  0  1
```

Reference vs Effect/Sum Coding

Sum or Effect Coding uses a contrast that compares each group to a “grand mean”.

The constraint is that the coefficients must sum to 0.

Reference vs Effect/Sum Coding

Back to the same example.

Using `RACE` with three categories as a predictor.

The sum or effect coding for this:

```
> contrasts(lowbwt$RACE)
      [,1] [,2]
1       1    0
2       0    1
3      -1   -1
```

Reference vs Effect/Sum Coding

The estimates from the example:

Reference Coding					Effect/Sum Coding				
RACE	RACE2	RACE3			RACE	RACE1	RACE2		
1	0	0			1	1	0		
2	1	0			2	0	1		
3	0	1			3	-1	-1		
	Estimate	Std. Error	z value	Pr(> z)		Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.155	0.2391	-4.83	1.36E-06 ***	(Intercept)	-0.6613	0.1759	-3.759	0.000171
RACE1	0				RACE1	-0.4937	0.2236	-2.207	0.02728
RACE2	0.8448	0.4634	1.823	0.0683 .	RACE2	0.3511	0.2889	1.215	0.224223
RACE3	0.6362	0.3478	1.829	0.0674	RACE3	0.1426			
Logit					Logit				
RACE=1	-1.155		-1.155	0.239577	RACE=1	-0.6613+(-0.4937)		-1.155	0.239577
RACE=2	-1.155+0.8448		-0.3102	0.423066	RACE=2	-0.6613 - 0.3511		-0.3102	0.423066
RACE=3	-1.155+0.6362		-0.5188	0.373133	RACE=3	-0.6613 - (-0.4937) - 0.3511		-0.5187	0.373156
					OR RACE=3	-0.6613+0.1426		-0.5187	

Model Fit

For two-way tables we considered the deviance, G^2 , which was a measure of how far the observed data departed from the expected data.

The saturated model has 0 for the deviance.

Model Fit

Recall that G^2 is based on the ratio of the likelihoods for the observed and expected data.

To apply that idea here, we need to start from the likelihood.

Model Fit

$$L(\beta) = \prod_{i=1}^n \pi_i^{Y_i} (1 - \pi_i)^{1-Y_i}$$

$$\log L(\beta) = \sum_{i=1}^n [Y_i \log \pi_i + (1 - Y_i) \log (1 - \pi_i)]$$

$$\log L(\beta) = \sum_{i=1}^n [Y_i \{\log \pi_i - \log(1 - \pi_i)\} + \log (1 - \pi_i)]$$

$$\log L(\beta) = \sum_{i=1}^n \left[Y_i \log \left(\frac{\pi_i}{1 - \pi_i} \right) + \log (1 - \pi_i) \right]$$

$$\log L(\beta) = \sum_{i=1}^n [Y_i \mathbf{X}'_i \beta + \log (1 - \pi_i)]$$

Model Fit

We can use this likelihood to test model fit.

First define $G_p = -2\log L(\hat{\beta})$ as the likelihood for the model with p coefficients.

Then define $G_k = -2\log L(\hat{\beta}_k)$ as a reduced model with k coefficients.

Then the likelihood ratio test can be written as $-2\log \frac{L(\hat{\beta}_k)}{L(\hat{\beta}_p)}$.

This ratio has a χ^2 distribution with $p - k$ degrees of freedom.

Model Fit

This can also be written as $G_k - G_p \sim \chi^2_{p-k}$.

This should allow us test a model vs an alternative.

The conclusion will be whether the full model contains additional information beyond the reduced model.

Model Fit

We can use this method to compare our hypothesized model to a null model.

The null model has only the intercept (the mean).

Do our covariates add anything to our knowledge of the probabilities?

We could also ask, can we successfully differentiate cases in our sample into those with different probabilities?

Model Fit

Let's go back to our O-ring example.

```
> anova(logitmod, test='LRT')
Analysis of Deviance Table

Model: binomial, link: logit

Response: damage

Terms added sequentially (first to last)

Df Deviance Resid. Df Resid. Dev Pr(>Chi)
NULL                                137      66.540
temp  1      6.144      136      60.396  0.01319 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

It appears that temperature adds to our understanding of the failure of O-rings.

Model Fit

Similar output directly from `glm` :

```
> logitmod
```

```
Call:  glm(formula = damage ~ temp, family = binomial,
```

```
Coefficients:
```

(Intercept)	temp
5.0850	-0.1156

```
Degrees of Freedom: 137 Total (i.e. Null); 136 Residu
```

```
Null Deviance: 66.54
```

```
Residual Deviance: 60.4 AIC: 64.4
```

Model Fit

This gives us a means to assess model fit and compare alternative models. We may also want to assess each coefficient in the model.

Fortunately, we can use some approximations here.

Model Fit

$\hat{\beta}$ is approximately distributed as $N_p(\beta, (\mathbf{X}'\mathbf{V}\mathbf{X})^{-1})$, where

$$\mathbf{V} = \text{diag}[\pi_1(1 - \pi_1), \pi_2(1 - \pi_2), \dots, \pi_n(1 - \pi_n)]$$

$$\mathbf{X} = \begin{pmatrix} \mathbf{X}'_1 \\ \mathbf{X}'_2 \\ \vdots \\ \mathbf{X}'_n \end{pmatrix}$$

Model Fit

```
> Anova(logitmod, type="II", test="Wald")  
Analysis of Deviance Table (Type II tests)
```

Response: damage

	Df	Chisq	Pr(>Chisq)
temp	1	6.0435	0.01396 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0

Model Fit

There is an assumed linear relationship between continuous predictors and the logit.

This assumption can be checked by reviewing *empirical logit plots*.

- 1 Calculate probability in binned values of the predictor.
- 2 Take the logit of these probabilities
- 3 Plot them. Is the relationship linear?

Model Fit

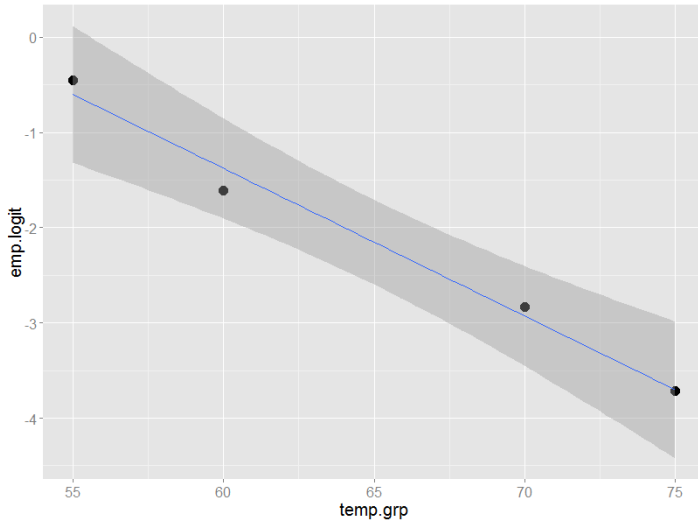
```
orings2<-orings
orings2$temp.grp[orings2$temp< 60] <- 55
orings2$temp.grp[orings2$temp >= 60 & orings2$temp <65 ] <- 60
orings2$temp.grp[orings2$temp >= 65 & orings2$temp <70 ] <- 65
orings2$temp.grp[orings2$temp >= 70 & orings2$temp <75 ] <- 70
orings2$temp.grp[orings2$temp>=75] <- 75

orings3<-summaryBy(damage ~ temp.grp, data = orings2,
  FUN = function(x) { c(sum = sum(x), cnt=length(x)) } )
orings3$p<-orings3$damage.sum/ (orings3$damage.cnt*6)
orings3$emp.logit<-log(orings3$p/ (1-orings3$p))
orings3<-orings3[-3,]
```

Model Fit

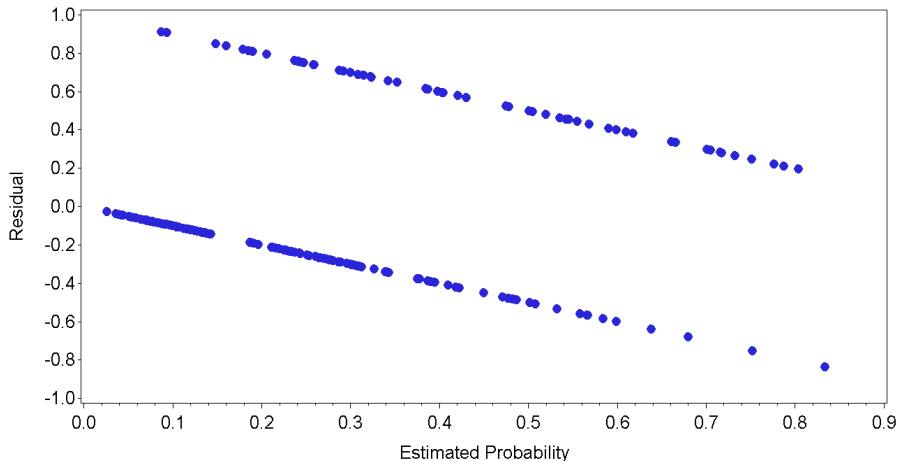
```
> orings3
temp.grp damage.sum damage.cnt      p emp.logit
1      55         7         3 0.38888889 -0.4519851
2      60         1         1 0.16666667 -1.6094379
4      70         2         6 0.05555556 -2.8332133
5      75         1         7 0.02380952 -3.7135721
```

Model Fit



Model Fit

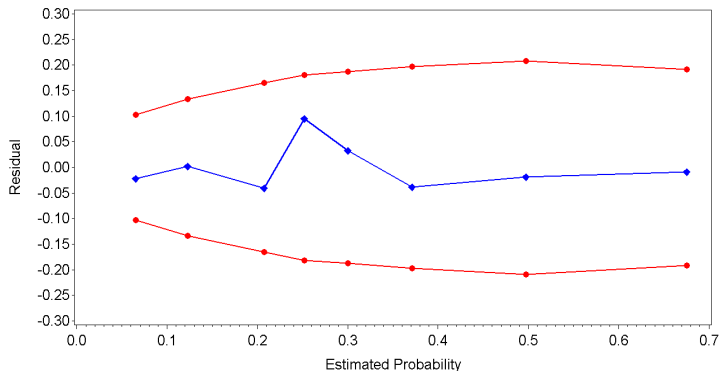
We can also look at the residuals. Here are the residuals plotted against the fitted values:



Model Fit

This is not very helpful as any given predicted value can only take on two values for the residual.

One method around this is to group cases, average their residuals and fitted values, and plot those. “Binned residual plot:”



Example 5

Table: Variables Available for Birth Weight Analysis

Variable	Description
ID	Identification Code
LOW	Low Birth Weight (0=Birth Weight greater than or equal to 2500g, 1=Birth Weight less than 2500g.)
AGE	Age of Mother in years
LWT	Weight in pounds at the last Menstrual Period
RACE	Race (1=White, 2=Black, 3=Other)
SMOKE	Smoking status During Pregnancy (1=Yes, 0=No)
PTL	History of Premature Labor (0=None, 1=One, etc.)
HT	History of Hypertension (1=Yes, 0=No)
UI	Presence of Uterine Irritability (1=Yes, 0=No)
FTV	Number of Physician Visits During the First Trimester (0=None, 1=One, 2=Two, etc.)
BWT	Birth Weight in grams.

Example 5

```
lowbwt.mod<-glm(LOW ~ AGE + LWT + factor(RACE) +  
    SMOKE + PTL + HT + UI + FTV,data=lowbwt)
```


Example 5

```
> lrtest(lowbwt.mod)
Likelihood ratio test

Model 1: LOW ~ AGE + LWT + factor(RACE) + SMOKE + PTL
Model 2: LOW ~ 1
#Df    LogLik Df  Chisq Pr(>Chisq)
1    11 -105.89
2     2 -122.80 -9  33.81  9.643e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0
```

Example 5

```
> roc(factor(lowbwt$LOW), lowbwt$phat, plot=TRUE, auc.print=TRUE)
Setting levels: control = 0, case = 1
Setting direction: controls < cases
```

Call:

```
roc.default(response = factor(lowbwt$LOW), predictor = lowbwt$phat, plot=TRUE)
```

Data: lowbwt\$phat in 130 controls (factor(lowbwt\$LOW) 0) < 59 cases (factor(lowbwt\$LOW) 1)

Area under the curve: 0.7469

Example 5

```
> PseudoR2(lowbwt.mod, which="all")
McFadden      McFaddenAdj      CoxSnell      Nagelkerke
0.1376655      0.0562313      0.1638022      0.2252136
AldrichNelson VeallZimmermann      Efron McKelveyZavoina
0.1517445      0.2685201      0.1638022      NA
Tjur      AIC      BIC      logLik
NA      233.7867436      269.4459608      -105.8933718
logLik0      G2
-122.7984909      33.8102383
```

These are various Pseudo- R^2 . It might be thought of as a standardized measure of improvement over the null model.

Not proportion of variance that is explained.

Example 5

$$\text{Nagelkerke Pseudo } R^2 = \frac{1 - \exp \left[-\frac{2}{n} \{ LL(M_{Full}) - LL(M_{Intercept}) \} \right]}{1 - \exp \left[-\frac{2}{n} LL(M_{Intercept}) \right]}$$

For example, in example 5, we have the estimated log-likelihoods (full and intercept-only):

$$\text{Pseudo } R^2 = \frac{1 - \exp \left[-\frac{2}{189} \{ -105.893 - -122.798 \} \right]}{1 - \exp \left[-\frac{2}{189} \times -122.798 \right]} = 0.2252$$

We might say that the covariates add about 23% more information beyond knowledge of the mean.

Example 5

Let's reduce the model. See R.

Example 5

```
> Anova(lowbwt.mod, type="II", test="Wald")
Analysis of Deviance Table (Type II tests)
```

Response: LOW

Df Chisq Pr(>Chisq)

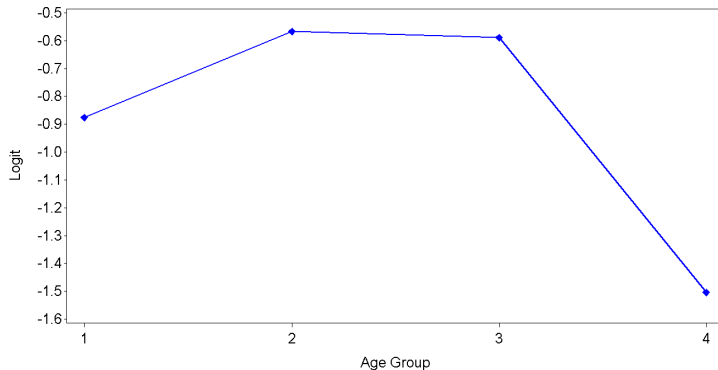
AGE	1	0.3296	0.565877
LWT	1	4.8142	0.028227 *
factor(RACE)	2	6.5050	0.038678 *
SMOKE	1	5.0295	0.024919 *
PTL	1	2.8770	0.089854 .
HT	1	7.3151	0.006838 **
UI	1	2.8347	0.092249 .
FTV	1	0.0413	0.838906

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Example 5

Is the relationship between AGE and LOW nonlinear?

Check the empirical logit plot:



Example 5

Try AGE and AGE*AGE:

```
> Anova(lowbwt.mod2, type="II", test="Wald")
Analysis of Deviance Table (Type II tests)
```

Response: LOW

Df	Chisq	Pr(>Chisq)
AGE	1 0.0280	0.867011
I (AGE^2)	1 0.0904	0.763636
LWT	1 3.4647	0.062692 .
RACE	1 3.9631	0.046509 *
SMOKE	1 5.1114	0.023769 *
PTL	1 2.8061	0.093906 .
HT	1 7.2650	0.007031 **
UI	1 2.7043	0.100078
FTV	1 0.0489	0.824986

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Example 5

Reduce the model:

```
> Anova(lowbwt.mod3, type="II", test="Wald")
Analysis of Deviance Table (Type II tests)
```

Response: LOW

```
Df  Chisq Pr(>Chisq)
```

LWT	1	4.7244	0.029738	*
RACE	1	5.0322	0.024881	*
SMOKE	1	7.3263	0.006795	**
HT	1	7.8663	0.005036	**
UI	1	4.3101	0.037887	*

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Example 5

Table: Comparing Two Models

Statistic	Model 2	Model 3
Log(L)	-107.110	-108.869
AIC	236.220	231.737
N-Pseudo R^2	0.210	0.188
AUC	0.746	0.728
p	9+Intercept	5+Intercept

Example 5

We can apply a test to the difference in the likelihoods, which will have a χ^2 distribution.

$G_6 - G_{10} = -2(-107.110 - -105.893) = 2.434$ with $9 - 5 = 4$ df. This is not significant at the 0.01 level since $\chi^2_{4,.01} = 13.2$.

Example 5

Same type of test can be used to test interactions in the model.

```
>lowbwt.mod3<-glm(LOW ~ LWT + RACE + SMOKE + HT + UI,data=lowbwt)
>lowbwt.mod4<-glm(LOW ~ LWT + RACE + SMOKE + HT + UI + SMOKE*LWT,
  data=lowbwt)
> lrtest(lowbwt.mod4,lowbwt.mod3)
Likelihood ratio test
```

Model 1: LOW ~ LWT + RACE + SMOKE + HT + UI + SMOKE * LWT

Model 2: LOW ~ LWT + RACE + SMOKE + HT + UI

#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	8	-108.81		
2	7	-108.87	-1 0.1202	0.7288